

The Mass Gap as a Tail Predicate

The Yang–Mills Verdict Is Not Frame-Internal

The circumscription obstruction at the continuum limit, the criticality dilemma, the corrected reading of the structural-gap argument, and the priced specification of the missing weld

Lu Semita · EmergenceByDesign Drawing on the author's Yang–Mills corpus (YANG-MILLS V2.1; the Structural–Constructive Outline; the Gribov–Singer material of The Dynamic Layer, Formalized, coauthored with Jennifer Lorraine Nielsen), the Semantic Obstruction calculus (SOIR; C1), the Residue Thesis (R1), and the companion papers Tail Predicates and the Circumscription Obstruction (T1, Navier–Stokes) and The Critical Line as a Tail Predicate (T2, Riemann), whose framework this paper instantiates at the continuum limit.

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Tier discipline throughout: **Tier I** is standard checkable mathematics, asserted with complete proof or standard citation; **Tier II** is exact structure read after the Tier-I objects are fixed; **Tier III** is declared direction. Falsification surfaces in §9.

0. Origin

This paper completes the triptych begun in T1 and T2, formalizing the same originating intuition at its third and sharpest station:

Each of these questions is of the same form — the way the question is posed is what generates the outcome. The asking precludes an answer by the means the asking suggests: one must arbitrarily stop imposing the same circumscription continuum to assert a definitive about what is unwitnessable from its own framing. The issue is not domain-dependent; it is structural.

Applied to Yang–Mills in the joint exchange that produced this triptych, the intuition took the form: *existence-plus-mass-gap is a construction demand whose finite witness — the constructed theory with certified spectrum — does not exist in any framing, and the lattice circumscription faces exactly the continuum-limit tail problem*. This paper proves that reading, and proves that Yang–Mills is the **sharpest** of the four instances — sharper than Navier–Stokes, Riemann, or Hodge — for a reason that will be made exact in §2: in the other domains the frame-internal stations are merely insufficient; here **every station of the circumscription continuum is completely and rigorously under control**, the lattice theory being fully constructed at every spacing, volume, and coupling — and the verdict nevertheless lives entirely, without remainder, in the limit the stations approach and never reach. Nothing is missing from the windows. The problem is the tail. No cleaner realization of the origin statement exists

in mathematics, and this paper's central new result — the **criticality dilemma** of Theorem C''' — shows that for Yang–Mills the two Clay clauses (existence; gap) are locked together by the tail structure itself: no station-wise gap statement can even be *formulated* as the Clay gap without the construction, and any gap mechanism uniform across stations refutes the existence it presupposes. The posing generates the outcome, here in its purest form.

The paper does not prove or refute existence or the gap. It proves what any proof must cost, corrects the author's own structural-gap argument to its indexed and defensible form (§6), and prices the missing weld — one component of which, uniquely among the four problems, has already been partially delivered *on Yang–Mills itself* (§7).

1. Setting

1.1 The Clay statement

For any compact simple gauge group G , prove that quantum Yang–Mills theory on $\mathbb{R}^4$ exists — a quantum field theory satisfying the Wightman axioms (or, equivalently via reconstruction, Euclidean Schwinger functions satisfying the Osterwalder–Schrader axioms) whose short-distance behavior is that of the classical theory — and that it has a mass gap $\Delta > 0$: the spectrum of the Hamiltonian in the vacuum sector is contained in $\{0\} \cup [\Delta, \infty)$.

The verdict is thus a conjunction: **EXIST** (a Σ -shaped demand: exhibit a construction satisfying Π -shaped axioms quantifying over all test functions and regions) **^** **GAP** (a spectral tail property of the constructed object's infinite-volume Hamiltonian).

1.2 The frames: a double circumscription continuum

Definition 1.2 (Yang–Mills frames). A *frame* is a station (a, L, g, n) of the following jointly ordered continua: Wilson lattice regularizations at spacing $a > 0$ and volume $L < \infty$ with bare coupling g ; perturbative expansions to order n ; numerical spectra at accessible parameters. A *window observable* is any functional of such a station.

The stations are completely rigorous (Tier I). This is the domain's distinguishing feature, and it must be stated at full strength:

1. **Wilson (1974):** lattice Yang–Mills at every (a, L, g) is a well-defined finite-dimensional integral over compact group Haar measures — not an approximation to a theory, but a complete rigorous statistical-mechanical object.
2. **Osterwalder–Seiler (1978):** the lattice theory satisfies reflection positivity at every station; at strong coupling (g large) the cluster expansion converges and yields, rigorously, exponential clustering — a lattice mass gap — and the Wilson-loop area law (confinement) at that station.
3. **Numerical lattice gauge theory:** glueball spectra, string tensions, and scaling behavior measured to high precision at accessible couplings.

And the verdict lives entirely in the tail (Tier I). The Clay theory is the double limit $a \rightarrow 0$ (continuum), $L \rightarrow \infty$ (infinite volume), taken along the trajectory dictated by asymptotic freedom (Gross–Wilczek, Politzer, 1973): the bare coupling must run to zero, $g(a) \rightarrow 0$, as $a \rightarrow 0$, at the rate fixed by the renormalization group, for the limit to have the required short-distance behavior. EXIST is the convergence of the Schwinger functions along this trajectory to a limit satisfying the OS axioms; GAP is a spectral property *of the limit*. Neither clause is a property of any station.

Definition 1.3 (Frame-internal decidability; weld). As in T1/T2 verbatim.

2. Classification: the purest tail predicate

Lemma 2.1 (Logical form; Tier I). EXIST is Σ -over-constructions of a Π -over-frames condition: the OS axioms for the limit quantify over all test functions, all reflection hyperplanes, all regions — verified by no finite frame. GAP is a tail predicate twice over: it concerns the spectrum of an operator that exists only after both limits, and spectral gaps are themselves infima over the full Hilbert space. Neither clause, nor their conjunction, is a window observable.

Proof. Quantifier inspection, as in T1 Lemma 2.1 and T2 Lemma 2.1; the only addition is that here even the *object* whose property is asserted (the continuum Hamiltonian) is frame-external — it does not exist at any station, only candidate approximants do. ■

Theorem 2.2 (Total window control, zero verdict information; Tier I/II). Every station of the Yang–Mills circumscription continuum is under complete rigorous control (§1.2, items 1–3), including a station-wise proven mass gap in the strong-coupling regime — and this complete control carries zero verdict information. The lattice gap of Osterwalder–Seiler is a gap *in lattice units at strong coupling*; the Clay gap is a gap *in physical units in the continuum limit at the asymptotically free trajectory's endpoint*; no theorem connects them, and Theorem C''' below shows the connection cannot be station-wise in principle. Yang–Mills thereby realizes the origin statement in its purest form: the frame-internal record is not merely unanimous but *complete*, and decides nothing.

Remark 2.3 (The window is treacherous here by theorem; Tier I). As in the arithmetic case (T2, Remark 2.3), the insufficiency of window extrapolation is exhibited *inside the domain* by a landmark theorem:

Triviality of ϕ^4_4 (Aizenman–Duminil-Copin, 2021; completing Aizenman/Fröhlich 1981–82). The four-dimensional lattice ϕ^4 theory is a well-defined, manifestly interacting statistical-mechanical object at *every* station (a, L, λ) — and its continuum scaling limits are Gaussian: free. Interaction at every window; freedom in the tail. This is the domain's Skewes phenomenon: a property unanimous across the entire circumscription continuum — interactivity — that the limit destroys. It is simultaneously the proof that the *negative* fate for a continuum-limit problem is achievable when a weld exists (the Gaussianity of the limit is an invariant, certified), and the standing warning that the existence half of the Clay problem is a genuine tail question: nothing about station-wise interactivity survives to the limit by default. The structural reason Yang–Mills is expected to escape ϕ^4_4 's fate — asymptotic freedom, the ultraviolet fixed point at zero coupling versus the quartic theory's Landau-pole growth —

is itself, at present, a perturbative (window-grade) fact whose nonperturbative promotion is precisely part of the missing weld.

3. The obstruction theorems

The counterparts of B/C/D (T1) and B'/C'/D' (T2). The domain-specific pillars: the triviality theorem, the abelian deconfinement theorem, and the renormalization-group scaling structure of the continuum limit.

Theorem B''' (Frame-blindness: the two counterfeits)

Theorem B''' (Tier I). Let $\mathfrak{F}$ be the *lattice frame*: the class of facts about a lattice field theory expressible from (i) well-definedness of the regularized theory at every station, (ii) reflection positivity, (iii) station-wise interactivity, (iv) station-wise strong-coupling confinement and clustering (area law, lattice gap), and (v) generic compact-group and correlation-inequality structure. Then:

1. **(Existence prong.)** No argument whose hypotheses are contained in (i)–(iii) + (v) can prove that the continuum limit of four-dimensional Yang–Mills exists as an interacting theory — because lattice ϕ_4^4 satisfies those hypotheses at every station and its continuum limit is free (Aizenman–Duminil-Copin). Any valid existence argument must engage structure beyond the generic lattice frame — in practice, asymptotic freedom made nonperturbative.
2. **(Gap prong.)** No argument whose hypotheses are contained in (i)–(v) can prove the continuum mass gap — because four-dimensional compact U(1) lattice gauge theory satisfies all of them, including rigorous strong-coupling confinement and lattice gap, and yet possesses a weak-coupling deconfined Coulomb phase with a massless photon (Guth 1980; Fröhlich–Spencer 1982, both rigorous): the gap present at every strong-coupling station does not survive passage toward the continuum regime. Any valid gap argument must engage the non-abelian structure *quantitatively along the trajectory*, not merely kinematically at stations.

Proof. Both prongs are counterfeit-witness arguments of the T1/T2 form: an exhibited object shares the frame facts and violates the conclusion; citations as stated. ■

Remark. Prong 2 vindicates, at theorem grade, the abelian/non-abelian contrast that the author's structural-gap corpus placed at its center (its "Abelian No-Go"): abelianness is the exempting structure, exactly as that corpus intuited — Guth and Fröhlich–Spencer are the rigorous certificate. What prong 2 simultaneously establishes is that invoking non-abelianness *kinematically* (holonomies fail to commute) cannot suffice, since the needed input is a coupling-dependent, trajectory-uniform mechanism. That sharpening is completed by the next theorem.

Theorem C''' (The criticality dilemma: windows buy windows, and the two clauses are locked)

Theorem C''' (Tier I/II). (i) The strong-coupling cluster expansion converges only for g above a threshold; the continuum trajectory requires $g \rightarrow 0$; no existing theorem carries the expansion's gap across the crossover: the

rigorous gap is a window fact, and windows buy windows. (ii) Perturbation theory is an asymptotic, generically divergent series: no order- n station decides existence, and the series does not define the theory. (iii) **(The di-lemma.)** The existence of the continuum limit with the required physics forces the lattice correlation length to diverge in lattice units along the trajectory: $\xi_{\text{lat}}(a) = \xi_{\text{phys}}/a \rightarrow \infty$ as $a \rightarrow 0$ — the theory must become *critical* in lattice units, its lattice-unit gap $\Delta_{\text{lat}} = a \cdot \Delta_{\text{phys}}$ tending to zero. Consequently:

1. Any structural argument yielding a lattice-unit gap bounded below *uniformly in the coupling* (hence uniformly along the trajectory) does not prove the Clay gap — it **refutes the Clay existence**, by forbidding the criticality that the continuum limit requires. Such an argument proves too much, and what it proves contradicts its own standing hypothesis when existence is assumed.
2. Any structural argument yielding only station-wise, coupling-dependent gaps proves nothing beyond Osterwalder–Seiler: a window fact.
3. Therefore the GAP clause admits no station-wise formulation at all: Δ_{phys} is defined only relative to the constructed limit, so the two Clay clauses are not separable targets — **the gap theorem cannot precede or bypass the existence theorem**, and any criterion strong enough to certify the gap along the trajectory (a uniform *physical*-unit spectral bound plus convergence of the Schwinger functions) is not a lemma toward the resolution but the resolution itself — the exact analogue of T1's Theorem C and T2's Theorem C'(ii).

Proof. (i) Osterwalder–Seiler convergence domain versus the asymptotic-freedom trajectory (Tier I). (ii) Classical (Dyson-type instability and large-order growth; the series' asymptotic character is standard). (iii) The scaling identity $\Delta_{\text{lat}} = a \cdot \Delta_{\text{phys}}$ is definitional once physical units are fixed by any renormalization condition; $a \rightarrow 0$ with Δ_{phys} finite forces $\Delta_{\text{lat}} \rightarrow 0$; branch 1 is the contrapositive; branch 2 is Theorem 2.2; branch 3 assembles them: a trajectory-uniform physical bound presupposes the trajectory's limit objects, i.e., the construction. ■

Theorem D''' (The escape routes)

Theorem D''' (Tier I/II). (i) (*Witness asymmetry.*) The positive verdict is Σ -witnessable: a construction, with OS axioms and a spectral bound certified, is a finite (presentable) witness — the fate achieved in lower dimensions (§4). The negative verdict — nonexistence of the interacting limit, or gaplessness — requires an invariant certified in the limit; the triviality theorem shows exactly what such a weld looks like when it exists (Gaussianity as the certified invariant), and no analogous invariant is known or indicated for Yang–Mills, where asymptotic freedom points the opposite way. (ii) (*The forcing lane, conditionally closed.*) Under the standard coding of the verdict — Schwinger functions as sequences of tempered distributions coded by reals, the OS axioms and the gap as projective conditions on that code — the statement sits at Σ^1_2 -type complexity, and Shoenfield absoluteness renders its truth value invariant under forcing; independence-via-forcing is then barred, exactly as at Hodge Door 4, with the condition (the coding's pointclass) declared rather than hidden. (iii) (*No cheap undecidability.*) A proof of "no ZFC construction exists" is a nonexistence claim over an open-ended construction space and requires the invariant of (i); pending one, the undecidability route is not an escape but a restatement of the negative fate at full price.

Proof. (i) is the classification of §4 plus the cited theorems. (ii) is the Shoenfield application, conditional as stated. (iii) is (i) restated at the metamathematical level. ■

Corollary F''' (The circumscription obstruction, continuum-limit form)

Corollary F''' (Tier II). The Yang–Mills verdict is not frame-internally decidable: not witnessed by any station (Lemma 2.1) despite total station-wise control (Theorem 2.2); not extrapolable from station facts, the domain's own landmark theorems exhibiting the failure of exactly that extrapolation on both clauses (Remark 2.3; Theorem B'''); not derivable from kinematic non-abelianness or any lattice-frame fact (B'''), nor from strong-coupling or perturbative windows (C''' i–ii); not formulable clause-by-clause, the gap being locked to the construction by the criticality dilemma (C''' iii); and not outflankable metamathematically at less than full price (D'''). Every decision factors through a weld whose hypothesis is trajectory-uniform, non-abelian in quantitative form, and convergence-carrying. The station at which one stops refining the lattice and pronounces is, absent such a weld, arbitrary. ■

4. The weld taxonomy, instantiated at the continuum limit

Type I — Criteria and transfer welds

- **The Osterwalder–Schrader reconstruction theorem (1973–75):** Euclidean Schwinger functions satisfying the OS axioms determine a Wightman quantum field theory, with the spectral condition — hence the meaningfulness of the gap — delivered by the reconstruction. **The transfer weld exists**, proven, exactly as the explicit formula does for Riemann: any candidate limit is tested against a fixed, completed coupling to the target structure. What is missing is the limit object, not the bridge.
- **Osterwalder–Seiler reflection positivity and the lattice cluster expansion:** the station-side partial welds — RP survives the limit if the limit exists (a structural inheritance criterion), and the strong-coupling expansion is the model of what a convergent control looks like, in the wrong regime.
- **Correlation inequalities and ϕ -bounds (Glimm–Jaffe):** the criteria toolkit that carried the lower-dimensional constructions.

Type II — Monotone and stability welds

- **Renormalization-group stability bounds:** Balaban's small-field/ large-field decompositions with inductively propagated bounds — the monotone-along-scales structure that substitutes, in constructive field theory, for an entropy functional: each RG step is controlled by the previous step's output, and the composite survives the scale limit.
- The two-dimensional c-theorem (Zamolodchikov) and its four-dimensional a-theorem descendant are noted (Tier II/III) as the field's entropy-weld archetypes; they order RG flows but do not, at present rigor, decide this problem.

Type III — Carrier welds: the constructions that already happened

The migration fate, documented within constructive quantum field theory — the exact counterpart of Weil/Deligne for RH and Markman for Hodge:

- **Glimm–Jaffe (and successors): ϕ^4_2 and ϕ^4_3** — interacting quantum field theories rigorously constructed, OS/Wightman axioms verified, **and the mass gap proven**, via cluster and phase-space expansions in which the continuum and infinite-volume tails become absolutely convergent series of finite objects. The full Clay-shaped conjunction — existence plus gap — has been *achieved*, in lower dimensions, by exactly the carrier this taxonomy names.
- **Gross–Neveu₂ (Feldman–Magnen–Rivasseau–Sénéor; Gawędzki–Kupiainen):** an **asymptotically free** theory rigorously constructed by multiscale RG expansion — the closest existing cousin of the Yang–Mills problem, and the proof that asymptotic freedom is not merely a hope but a *usable mechanism* for nonperturbative construction.
- **Balaban (1984–89): ultraviolet stability for four-dimensional lattice Yang–Mills in finite volume** — uniform bounds along the RG trajectory for the gauge theory itself. This is a partial delivery of the carrier *on the Clay problem's own object*: the UV half of (W2'') below, in finite volume, exists. (Magnen–Rivasseau–Sénéor obtained related results by different methods.) No other Millennium problem in this triptych possesses a partially built weld of this depth.
- **The negative-side exhibit:** Aizenman–Duminil-Copin's triviality theorem — the nonexistence weld executed, certifying the Gaussian invariant in the limit — the proof that the taxonomy's negative fate is real and what its certificate looks like.

5. The spine, completed

The four-column table of T2 §5 extends to its final row-set; the Yang–Mills column reads: two coupled ladders — the bare-coupling/lattice ladder and the physical-spectrum ladder, coupled by the renormalization group; verdict's logical form — Σ -over-constructions of Π -axioms, conjoined with a spectral tail of the limit object; frame-internal bar — Theorems B'', C'', D'' (triviality counterfeit, abelian counterfeit, the criticality dilemma); window record, verdict-silent — *complete* rigorous control at every station including a strong-coupling gap, plus high-precision numerics; weld that succeeded / is specified — Glimm–Jaffe and Gross–Neveu₂ (the conjunction achieved in lower dimension; asymptotic freedom used constructively), Balaban (the UV carrier partially built on Yang–Mills itself); specification (W1'')–(W4'').

The general schema, third confirmation (Tier II). The seam here is the renormalization-group coupling of the discrete regulated description to the continuum spectral one; the verdict is an exact global property of the seam's limit; the native framing offers only station refinement; the domain's own theorems bar every frame-internal route; and every recorded decision — every constructed theory, and the one proven triviality — is a carrier enlargement. The outcome is generated by the posing, for the third time, by the same structure.

The prime-origin closure (Tier II). Yang–Mills gives the home-domain contrast its extreme form. In arithmetic, each instance is decidable while the aggregate is tail; here each station is not merely decidable but a *completely constructed theory* — richer than any single fact — and the aggregate is still tail, entirely. The Rope-and-Sand lesson terminates here: no accumulation of perfected windows, however perfect, ever amounts to the limit; the limit is a different kind of object, reached only by a weld that makes it one.

6. The corrected moves: the structural-gap corpus, replayed

The author's Yang–Mills corpus (the *Structural Proof of the Yang–Mills Mass Gap* and its outline; the Gribov–Singer material of the coauthored *Dynamic Layer*) is replayed minus the identified error, each move landing on a defensible station (Tier II ledger):

1. **The scope discipline is retained and credited.** The corpus states explicitly that it does not construct the continuum, does not compute the gap, and asserts only a conditional implication given existence. This is a genuine advance in posture over the coauthored NS and RH drafts, and it is exactly the traversal discipline; the correction below concerns the implication's content, not its framing.
2. **The abelian/non-abelian contrast is retained and promoted.** The corpus's Abelian No-Go — commuting holonomies admit global sections and exempt abelian theories — is vindicated at theorem grade by Theorem B'' prong 2: Guth and Fröhlich–Spencer are the rigorous certificate that abelianness is the exempting structure. The corpus's central instinct is correct and is now anchored to the classical literature that proves it.
3. **The Gribov–Singer material is retained as the geometric face.** Singer's theorem (1978) — no continuous global gauge fixing on the configuration space of non-abelian connections — is Tier I topology, and the corpus's use of it as the signature of irreducible non-abelian structure is sound. It is retained here as the candidate *geometric language* for the (W3'') mechanism: the horizon structure of the quotient space is where a trajectory-uniform, intrinsically non-abelian coercivity would naturally live.
4. **The central implication is corrected at the criticality joint.** The corpus's chain — irreducible commutator cycles (Theorem 3.1) $\implies$ structural area law $\implies$ gap via OS reconstruction — draws its premise from kinematic algebra that holds *uniformly in the coupling*: the non-commutativity of holonomies is as true at $g = 10^{-6}$ as at $g = 10$. By Theorem C'', a premise uniform along the trajectory whose asserted consequence is a gap must yield a lattice-unit gap uniform along the trajectory — which refutes the criticality that the corpus's own standing hypothesis (existence of the continuum theory) requires. The dilemma is exact: read as trajectory-uniform, the implication proves too much and contradicts its hypothesis; read as station-wise, it re-derives Osterwalder–Seiler's strong-coupling facts — a window statement in tail dress. The honest residue is substantial: the corpus has correctly identified the *mechanism candidate* — quantitative non-abelian commutator structure — whose still-missing, coupling-dependent, trajectory-uniform form is precisely component (W3'') of §7. The intuition located the right door; the implication mistook the door's address for its key.

5. **Scheme independence is corrected to its dependency.** The corpus's §6 argues the gap $m_{\mathcal{S}} = c_{\mathcal{S}} \Lambda_{\mathcal{S}} > 0$ in every admissible scheme; but $\Lambda_{\mathcal{S}} > 0$ — the existence of a finite dynamically generated scale — presupposes the constructed nonperturbative limit. The scheme-independence observation is correct *conditional on* (W2'') and is retained as such: it belongs after the construction, not before.

Nothing of value is discarded: the posture, the contrast, the geometry, and the mechanism candidate are all retained at their indexed strength; one implication is replaced by the theorem that shows what it must become.

7. The priced specification: what a proof must produce

By Corollary F''' and §4, a proof of the Clay conjunction must be a Type III carrier weld scaffolded by Type II stability, its hypothesis trajectory-uniform and quantitatively non-abelian. The mandatory components, each priced by the exhibit showing it achievable:

(W1''') The carrier. The multiscale renormalization-group phase-space expansion along the asymptotically free trajectory — the framework in which the double tail ($a \rightarrow 0, L \rightarrow \infty$) becomes a convergent composition of controlled finite steps. *Priced by:* Balaban's ultraviolet stability for lattice Yang–Mills in finite volume — the carrier partially built, on this problem's own object; and by Gross–Neveu₂, where an asymptotically free trajectory was traversed to completion.

(W2''') Existence: uniform bounds and the double limit. Propagation of Balaban-type stability to full uniformity (all scales, infinite volume), tightness and convergence of the Schwinger functions along the trajectory, and verification of the OS axioms for the limit. *Priced by:* Glimm–Jaffe φ_2^4/φ_3^4 (the conjunction completed in lower dimension) and the Feldman–Magnen–Rivasseau–Sénéor / Gawędzki–Kupiainen constructions.

(W3''') The gap mechanism, trajectory-uniform and non-abelian. A nonperturbative spectral lower bound — equivalently a string-tension or clustering bound — established *in physical units, uniformly along the trajectory*, surviving both limits; necessarily engaging non-abelian structure beyond kinematics (Theorem B''' prong 2) and necessarily coupling-dependent in lattice units (Theorem C'''). Candidate geometric faces: center-vortex and monopole condensation mechanisms made rigorous; the Gribov–Singer horizon geometry of §6 item 3. *Priced by:* the mass-gap halves of the φ_2^4/φ_3^4 constructions (a gap carried through a continuum limit, executed), with the abelian deconfinement theorem standing as the control that fixes what the mechanism must use.

(W4''') The transfer. Osterwalder–Schrader reconstruction — **already proven**. The limit's OS data yield the Wightman theory, Hamiltonian, and spectral condition without further work.

The inversion pattern of the triptych completes symmetrically: for Navier–Stokes the transfer was cheap and the profile is missing; for Riemann the transfer is proven and the carrier is missing; for Yang–Mills the transfer is proven **and the carrier is partially built** — Balaban's finite-volume UV stability is a delivered fragment of (W1''')/(W2''') that neither of the other problems can match. Necessity, relative to §3: without (W1''') the double tail is uncompactified (Lemma 2.1); without (W3''')'s uniformity the criticality dilemma revives; without its non-abelian content the abelian counterfeit revives; (W4''') exists. **This is the missing weld, stated as a work order —**

and, as with Riemann, it is demonstrably the work order the constructive program has been executing, which is the structural reading's third and final confirmation.

8. What this paper proves, in one paragraph

That the Yang–Mills verdict is a conjunction of a Σ -shaped construction demand and a spectral tail property, witnessed by no station of a doubly infinite circumscription continuum (Lemma 2.1) — while, uniquely among the Millennium problems treated in this triptych, *every* station is under complete rigorous control, so that the problem is purely and without remainder the tail (Theorem 2.2); that window extrapolation fails here by the domain's own landmark theorems — interactivity destroyed in the ϕ_4^4 limit, the strong-coupling gap lost in the abelian weak-coupling phase — yielding a two-pronged frame-blindness theorem: existence arguments must exceed the generic lattice frame, gap arguments must exceed kinematic non-abelianness (Theorem B'''); that the criticality dilemma locks the two Clay clauses together — a trajectory-uniform lattice-unit gap refutes the existence it presupposes, a station-wise gap restates Osterwalder–Seiler, and the physical gap is not even formulable without the construction, so no gap theorem can precede the existence theorem (Theorem C'''); that the escape routes are priced — the negative fate requires a triviality-theorem-grade invariant that does not exist for Yang–Mills, and the forcing lane is conditionally closed by absoluteness (Theorem D'''); that the author's structural-gap corpus, replayed through these theorems, retains its posture, its abelian/non-abelian contrast (now theorem-anchored), its Gribov–Singer geometry, and its mechanism candidate, with exactly one implication corrected at the criticality joint (§6); and that a proof must consist of the multiscale carrier, the uniform double-limit bounds, and the trajectory-uniform non-abelian gap mechanism, added to the already-proven Osterwalder–Schrader transfer — a specification one component of which, Balaban's ultraviolet stability, is already partially delivered on Yang–Mills itself (§7).

9. Standing and falsification surfaces

Asserted as proven: Lemma 2.1; Theorems 2.2, B''', C'''(i)–(iii), D'''(i)–(iii) at the tiers marked, with proofs or standard citations as given; Corollary F''', the taxonomy classifications, the spine completion, and the necessity direction of §7 (Tier II relative to the Tier-I rows).

Not asserted: EXIST, GAP, their negations, their independence, or the achievability of (W1''')–(W3''') in four dimensions (achieved in lower dimensions and for Gross–Neveu₂; partially achieved for Yang–Mills by Balaban; open as a whole); any claim that the structural-gap corpus's *intuition* was unsound — §6 argues the contrary.

To defeat this paper, exhibit one of:

- **F1'''** — an error in the proof of 2.2, B''', C''', or D''';
- **F2'''** — a proof of continuum existence from generic-lattice-frame hypotheses alone (refutes B''' prong 1 by refuting the triviality theorem's counterfeit role), or a proof of the continuum gap from kinematic non-abelianness plus lattice-frame facts alone (refutes prong 2 by refuting the abelian deconfinement counterfeit);

- **F3'''** — a trajectory-uniform lattice-unit gap theorem *consistent* with the existence of the continuum limit (dissolves the criticality dilemma);
- **F4'''** — a rigorous decision of a continuum-limit verdict in quantum field theory outside the three-type taxonomy;
- **F5'''** — a valid resolution of the Clay conjunction lacking (W1''') or (W3''').

Absent such an exhibit, the origin statement stands as proven at its third station and at exactly its earned strength: *the way the question is posed is what generates the outcome — the Yang–Mills verdict is not frame-internal; the circumscription continuum, though completely mastered at every station, decides nothing at any of them; the two clauses of the problem are locked together by the tail structure itself; and every road to the mass gap runs through a carrier whose specification is complete, whose transfer is proven, and whose foundation — alone among these problems — is already partially laid.*

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